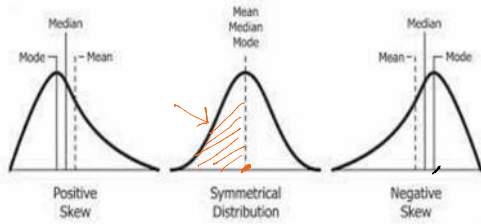


positively skew distribution

— longer tail towards the higher values of the variable

negatively skew distribution —
longer tail towards the lower values of the variable

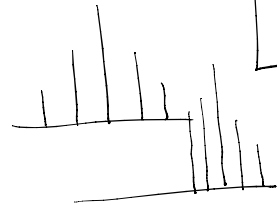


positively skew
 $\text{mode} < \text{median} < \text{mean}$

negatively skew
 $\text{mean} < \text{median} < \text{mode}$

mode is the value where frequency / frequency density is highest

m_3 — 3rd central moment
 $m_3 = 0$ for symmetrical distribution
 $m_3 > 0$ for positively skew
 $m_3 < 0$ for negatively skew.



(~~every~~ all odd order central moments are zero for symmetrical distribution).

Pearson's β_1 coefficient

$$\beta_1 = \frac{m_3}{m_2^{3/2}} = 0 \quad \text{if not symmetrical}$$

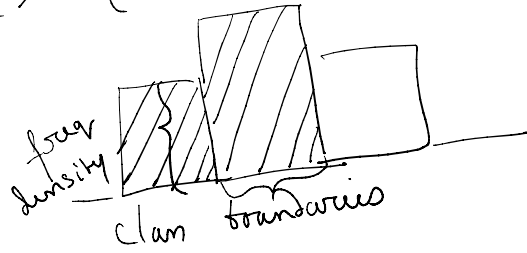
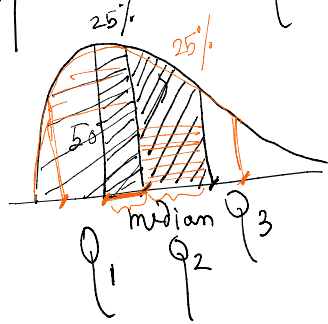
$$\gamma_1 = \frac{m_3}{m_2^{3/2}} = 0 \quad \text{for symmetrical dist}$$

$$> 0 \quad \text{for +vely skew}$$

$$< 0 \quad \text{for -vely skew}$$

$$Q_2 - Q_1 > Q_3 - Q_2$$

for +vely skew $Q_3 - Q_2 > Q_2 - Q_1$

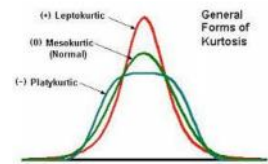


for -vely skew $Q_3 - Q_2 < Q_2 - Q_1$

$$Q = \frac{Q_3 - Q_1}{2} \quad (\text{quartile deviation or semi-interquartile range}).$$

Kurtosis

high kurtosis — more pointed and has thick tails compared to a normal distribution with same variance. (high concentration of values near central tendency)



low kurtosis — flat curve and has thin tails compared to a normal distribution with same variance. (low concentration of values near central tendency)

Pearson's β_1, γ coefficient

$$\beta_2 = \frac{m_4}{m_2^2}$$

$$\beta_2 = 3$$

$$> 3$$

$$< 3$$

for normal curve

for +ve kurtosis

for -ve kurtosis

for normal curve
kurtosis

$$\gamma_2 = \beta_2 - 3 = \begin{cases} < 0 & \text{for normal curve} \\ = 0 & \text{for +ve kurtosis} \\ > 0 & \text{for -ve kurtosis} \end{cases}$$

Ex 1. The sd calculated from a set of 32 observations is 5. If the sum of the observations is 80, what is the sum of the squares of these observations.

Ex 2. The mean and sd of 20 items is found to be 10 and 2 respectively. One item whose value is 8 replaced by an item with value 12. What will be the new mean and sd?